CHAPTER 5

THE SMALL SIGNAL LOW FREQUENCY ANALYSIS MODEL OF BJT

The modelling of BJT when only AC signal is applied is called small signal modelling of BJT. It is used to analyse the amplifier ckt or BJT ckt in its various configurations by calculating i/p and o/p impedance, voltage and current gain.

These characteristics are determined by using β and ckt resistance value.

It is used to conduct

Small signal re (T) model: In small signal re-model, the equivalent parameters of ckt are determined by the actual operating conditions rather than by using a data sheet value.

AC and DC equival. ckt:

DC equivalent circuit - To convert the ckt into DC equivalent form, following steps must be remembered-

1. Reduce AC sources to zero.
2. Replace capacitors by open ckt.

AC equivalent ckt -

1. Reduce all DC sources to zero.
2. Replace capacitors with short ckt.

$I_E = I_C + I_B$ $\qquad I_C = \beta I_B$

$= \beta I_B + I_B$

$= \cancel{\beta} I_B(\beta+1) \approx I_B \beta$

$\therefore I_E \simeq I_C$

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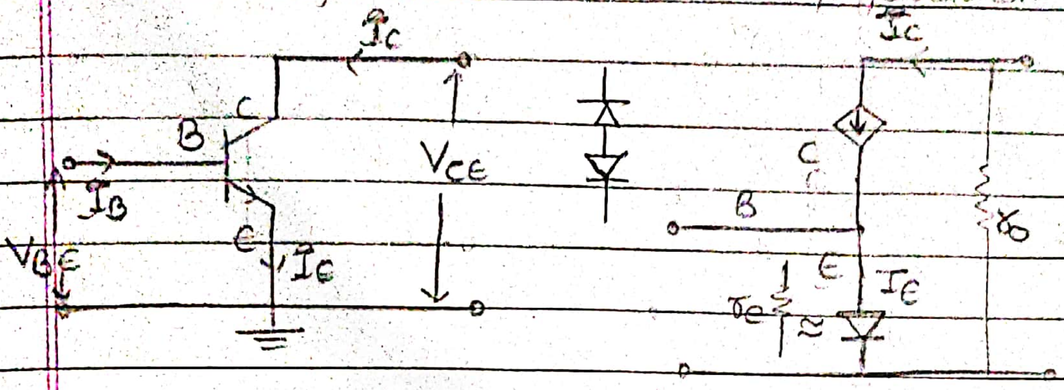
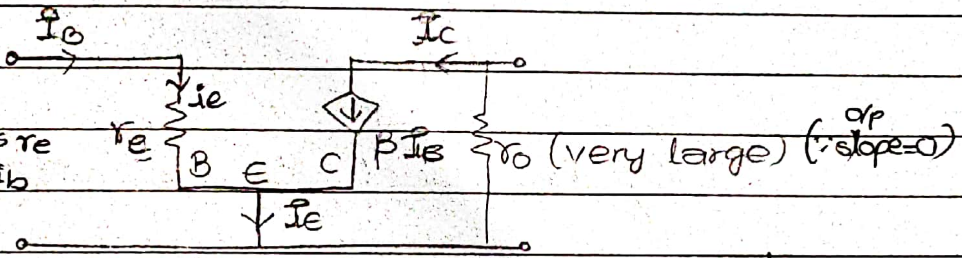
re model of common emitter configuration





drop across r_e

$= i_e \cdot r_e = (\beta+1) r_e i_b$

$\approx \beta r_e i_b$

$I_e = i_c + i_b$

$i_e = i_b(\beta+1)$

From diode current eq^n, we have,

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right)$$

$$\Rightarrow \frac{dI_D}{dV_D} = \frac{I_S}{\cancel{\eta} V_T} e^{\frac{V_D}{\cancel{\eta} V_T}} \qquad (\because \eta = 1)$$

$$\Rightarrow \frac{1}{r_d} = \frac{I_D + \cancel{I_S}}{V_T}$$

$$\Rightarrow \frac{1}{r_d} = \frac{I_D}{V_T} \qquad \therefore r_d = \frac{26\,mV}{I_D} \quad (V_T = 26mV)$$

AC →

$$\boxed{\therefore r_e = \frac{26mV}{I_E}}$$

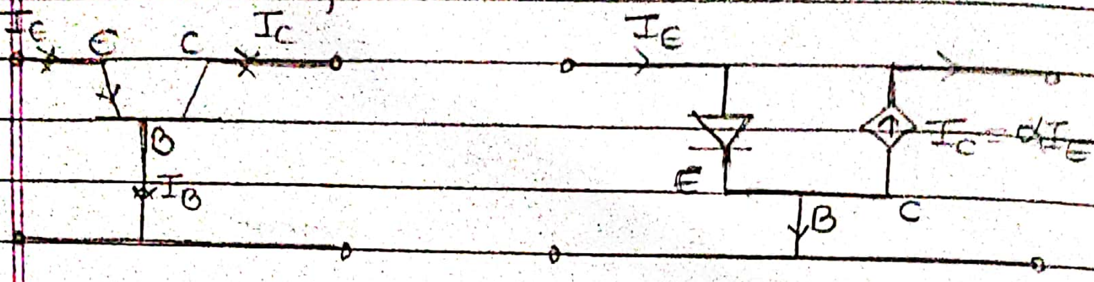
1. I/p resistance (Z_i) $= \cancel{\frac{26mV}{I_E}} \frac{V_{BE}}{I_B} = \frac{V_{BE} \times \cancel{I_E}}{\frac{I_E}{(\beta+1)}} = r_e\beta$

2. O/p resistance (Z_o) $= \frac{V_{CE}}{I_C} = \frac{I_C \times r_o}{I_C} = r_o$ (* in CE r_o is consid.)

3. Voltage gain (A_v) $= \frac{V_o}{V_i} = \frac{r_o \cdot \beta I_B}{r_e \cdot \beta I_B} = \frac{r_o}{r_e}$

4. Current gain (A_i) $= \frac{I_o}{I_i} = +\frac{I_c}{I_B} = +\frac{\beta I_B}{I_B} = +\beta$

r_e model of CB BJT :



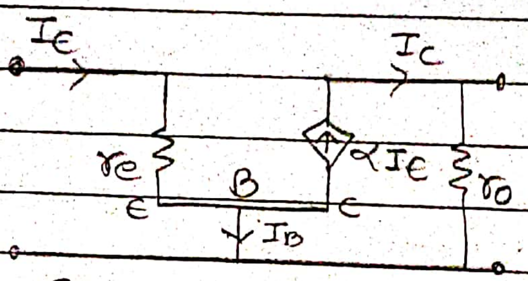


fig- re model for CB configuration

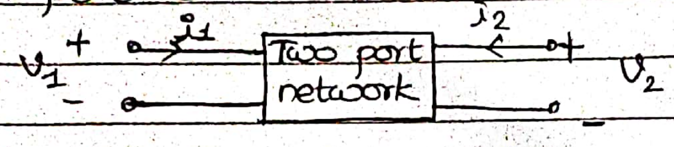
1. Input resistance $(Z_i) = \frac{\Delta V_{BE}}{\Delta I_E} = \frac{I_E \times r_e}{I_E} = r_e$

2. Output resistance $(Z_o) = \frac{\Delta V_{CB}}{\Delta I_C} = \frac{I_C \times r_o}{I_C} = r_o \simeq \infty$

3. Current gain $(A_i) = \frac{I_C}{I_E} = \frac{\alpha I_E}{I_E} = \alpha$

4. Voltage gain $(A_v) = \frac{V_{CE}}{V_{CB}} = \frac{I_C \cdot r_o}{I_E \cdot r_e} = \frac{\beta I_B r_o}{\beta I_B r_e} = \frac{r_o}{r_e}$

h-parameter ~~A-model~~ model- Another modeling of transistor makes use of two port network structure with four different parameters called hybrid or h-parameters.



$$v_1 = f(i_1, v_2)$$
$$i_2 = f(i_1, v_2)$$

For small signal operation, change in v_1 and i_2 may be expressed as,

$$dv_1 = \frac{\partial v_1}{\partial i_1}\partial i_1 + \frac{\partial v_1}{\partial v_2}\partial v_2$$

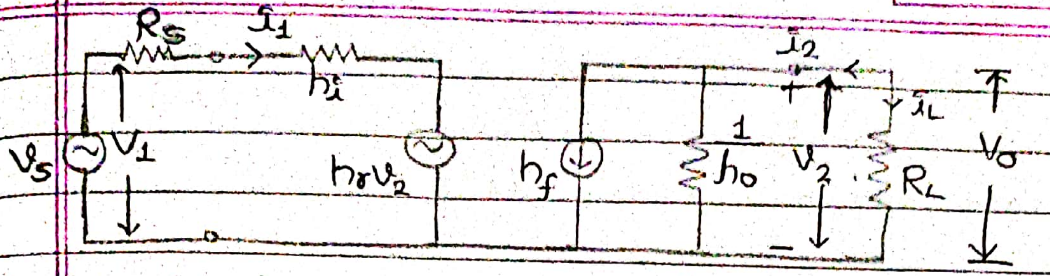
$$\therefore di_2 = \frac{\partial i_2}{\partial i_1}\partial i_1 + \frac{\partial i_2}{\partial v_2}\partial v_2$$

$$\therefore \boxed{\begin{aligned} v_1 &= h_{11} i_1 + h_{12} v_2 \\ i_2 &= h_{21} i_1 + h_{22} v_2 \end{aligned}} \quad \begin{aligned} &\cdots (1) \\ &\cdots (2) \end{aligned}$$

1. When $v_2 = 0$; i) $v_1 = h_{11} i_1$ $\therefore h_{11} = \frac{v_1}{i_1}$ (I/p impedence)
   ii) $h_{21} = \frac{i_2}{i_1}$ (current gain)
2. When $i_1 = 0$:
   i) $v_1 = h_{12} v_2$ $\Rightarrow h_{12} = \frac{v_1}{v_2}$ (voltage ratio)
   ii) $h_{22} = \frac{i_2}{v_2}$ $\Rightarrow$ (O/p admittance)

Now, eq^n (1) & (2) can be written as,

$$v_1 = h_i i_1 + h_r v_2 \quad --(3)$$
$$i_2 = h_f i_1 + h_o v_2 \quad --(4)$$



1. Voltage gain:- We know, $A_v = \dfrac{\text{o/p voltage}}{\text{i/p voltage}} = \dfrac{V_2}{V_1}$

From ckt,

$V_0 = V_2 = i_L R_L = -i_2 R_L$ ---- (a)

Also, $i_2 = -\dfrac{V_2}{R_L}$

From eqⁿ (4), $-\dfrac{V_2}{R_L} = h_f i_1 + h_o V_2$

$\Rightarrow -V_2\left[\dfrac{1}{R_L} + h_o\right] = h_f i_1$

$\Rightarrow i_1 = -\dfrac{V_2}{R_L h_f}\left[\dfrac{1}{R_L} + h_o\right]$ ---- (b)

Substituting value of i_1 in (1),

$V_1 = -\dfrac{h_i V_2}{h_f}\left[\dfrac{1}{R_L} + h_o\right] + h_r \cdot \cancel{(-i_2 R_L)}\, V_2$

$\Rightarrow V_1 = V_2\left\{\dfrac{-h_i(1+h_o R_L)}{h_f R_L} + h_r\right\}$

$\Rightarrow V_1 = V_2\left(\dfrac{-h_i(1+h_o R_L) + h_r h_f R_L}{h_f R_L}\right)$

$$\boxed{\therefore \frac{V_2}{V_1} = \frac{-h_f R_L}{-h_i(1+h_o R_L) + h_r}}$$

NOTE: Negative sign indicates that voltage is 180° out of phase.

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2. Current gain (A_i): We know, $A_i = \frac{i_L}{i_1} = \frac{-i_2}{i_1}$

From ckt, $V_0 = V_2 = -i_2 R_L$

From eq^n (2), $i_2 = h_f i_1 + h_o(-i_2 R_L)$

$\Rightarrow i_2 = h_f i_1 - h_o R_L i_2$

$\Rightarrow i_2(1 + h_o R_L) = h_f i_1 \Rightarrow \frac{i_2}{i_1} = \frac{h_f}{1 + h_o R_L}$

$$\boxed{\therefore A_i = \frac{-i_2}{i_1} = \frac{-h_f}{1 + h_o R_L}}$$

3. Input impedance (Z_{in}):

Here, $Z_{in} = \frac{V_1}{i_1}$

From (4), $\frac{-V_2}{R_L} = h_f i_1 + h_o V_2$

$\Rightarrow -V_2\left(\frac{1}{R_L} + h_o\right) = h_f i_1$

$\therefore V_2 = \frac{-h_f i_1 R_L}{1 + h_o R_L}$

Now, from (1), $V_1 = h_i i_1 + h_r V_2$

$\Rightarrow V_1 = h_i i_1 - \frac{h_r h_f i_1 R_L}{1 + h_o R_L}$

$\Rightarrow V_1 = i_1\left(h_i - \frac{h_r h_f R_L}{1 + h_o R_L}\right)$

$$\boxed{\therefore Z_{in} = \frac{V_1}{i_1} = \left(h_i - \frac{h_r h_f R_L}{1 + h_o R_L}\right)}$$

4. Output impedance (Z_o):

Here, $Z_o = \frac{V_2}{i_2}\Big|_{V_s = 0}$

Applying KVL in i/p loop,

$i_1 R_s + i_1 h_i + h_r V_2 = 0$

$\therefore i_1 = \frac{-V_2 h_r}{R_s + h_i}$

From (2), $i_2 = h_f \cdot \left(\frac{-v_2 h_r}{R_s + h_i}\right) + h_o v_2$

$\Rightarrow i_2 = v_2 \left(\left(\frac{-h_f h_r}{h_i + R_s}\right) + h_o \right)$

$\Rightarrow \frac{v_2}{i_2} = \frac{h_i + R_s}{(h_o h_i + h_o R_s - h_f h_r)}$

$\therefore Z_o = \frac{h_i + R_s}{(h_o h_i + h_o R_s - h_f h_r)}$

Hybrid-π model-

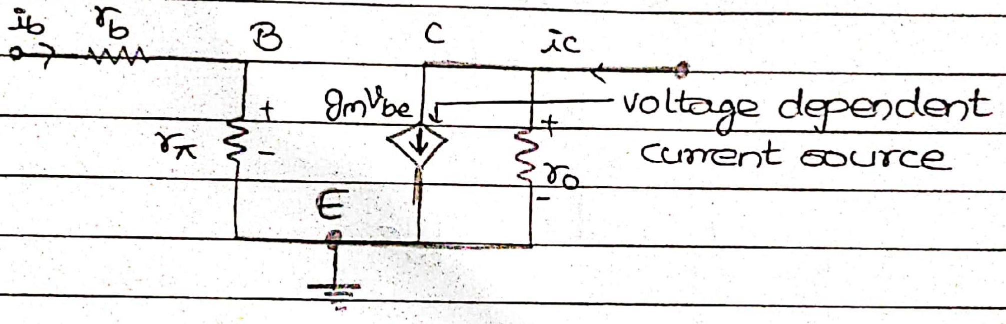


Fig- simplified hybrid-π model

Since hybrid-π model uses g_m, it is often called transconductance model.

Here, $g_m = \left.\frac{i_o}{V_i}\right|_{V_o=0} = \left.\frac{I_c}{V_{be}}\right|_{V_{ce}=0} = \frac{i_c}{26mV}$

$\Rightarrow g_m = \frac{i_e}{26mV} \quad (\because i_c \approx I_e)$

$\therefore g_m = \frac{1}{r_e}$

And, $r_\pi = \beta r_e$

$\Rightarrow r_\pi = \frac{\beta}{g_m}$

$\therefore \beta = g_m r_\pi$

$\cancel{r_\pi} = \frac{\cancel{V_{be}}}{\cancel{I_{be}}}$

$\cancel{r_\pi} =$

$$\boxed{\begin{aligned} I_C &= I_S\left(e^{\frac{V_{be}}{\eta V_T}} - 1\right) \\ \Rightarrow \frac{\partial I_C}{\partial V_{be}} &= \frac{I_C + \cancel{I_S}^{\,0}}{V_T} \\ \therefore g_m &= \frac{I_C}{26mV} \end{aligned}}$$

* if early effect is negligible
then $V_A = \infty$
$r_o = \infty$

$$g_m \qquad \beta$$

1. $Z_{in} = \dfrac{\Delta V_{be}}{\Delta i_b} = \dfrac{i_b \cdot r_\pi}{i_b} = r_\pi = \left(\dfrac{V_{be}}{i_c}\right)\left(\dfrac{i_c}{i_b}\right) = \dfrac{\beta}{g_m} = \dfrac{V_T \beta}{i_c}$

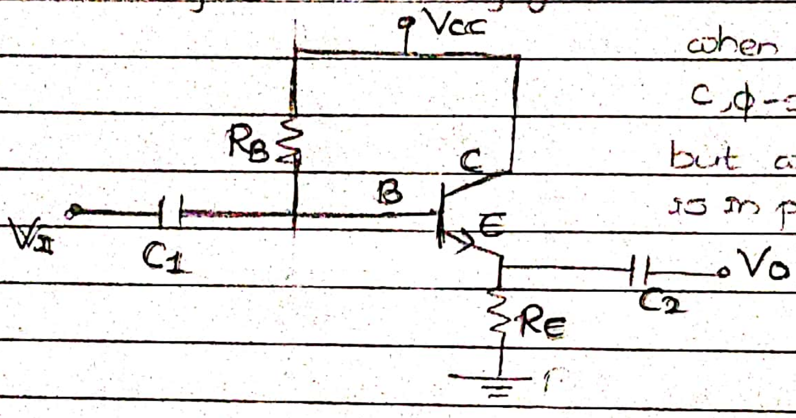
$$\boxed{\Rightarrow r_\pi = \frac{V_T \beta}{\beta i_b} = \frac{V_T}{i_b}}$$

2. $Z_o = \dfrac{\Delta V_{ce}}{\Delta i_c} = \dfrac{i_c \cdot r_o}{i_c} = r_o = \dfrac{V_A + V_{ce}}{i_c}$ — early voltage

3. $A_i = \dfrac{\Delta i_c}{\Delta i_b} = \dfrac{g_m V_{be}}{i_b} = g_m r_\pi$

4. $A_v = \dfrac{\Delta V_{ce}}{\Delta V_{be}} = \dfrac{-g_m V_{be} r_o}{V_{be}} = -g_m r_o$

Emitter follower configuration–

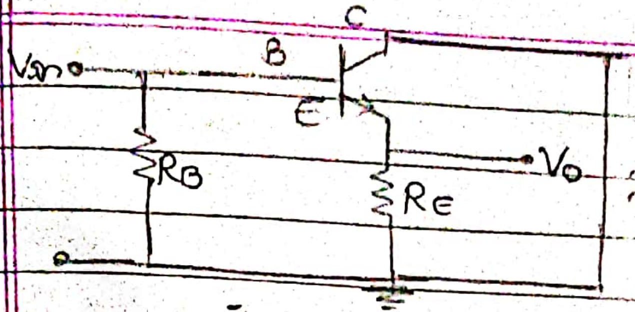


when o/p taken from C, ϕ-shift → 180°
but when from E, V_I is in phase with V_o so called emitter follower.

fig- Emitter follower configuration

When o/p is taken from collector, it has 180° phase shift betn o/p and i/p signal. When o/p is taken from emitter o/p is in phase with i/p. So, called emitter follower configuration.

- It is used for impedance matching
- It provides high i/p impedance and low o/p impedance.



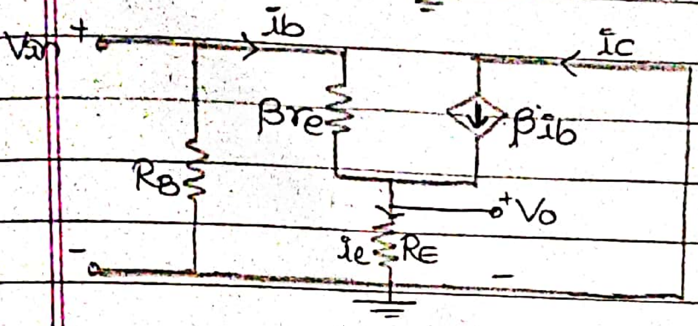


fig- re equivalent model

$i_e = i_b + i_c$
$= (\beta+1) i_b$
$V_e = \beta i_b R_E$

1) Here, $Z_i = R_B \| (\beta r_e + R_E)$

$= R_B \| (\beta r_e + \beta R_E)$

$\therefore Z_i = R_B \| \beta R_E \qquad (\because R_E >>> r_e)$

At first,

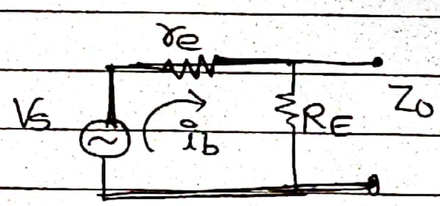
$$i_b = \frac{V_{in}}{\beta r_e + \beta R_E}$$

$$i_e = \beta i_b = \frac{\beta V_{in}}{\beta (r_e + R_E)} = \frac{V_{in}}{r_e + R_E}$$

2) Now, $Z_o = r_e \| R_E$

$$= \frac{r_e \times R_E}{r_e + R_E} = \frac{r_e R_E}{R_E}$$

$\therefore Z_o = r_e$.



3) Voltage gain $(A_v) = \frac{V_o}{V_i} = \frac{i_e R_E}{V_{in}}$

$$= \frac{V_{in} R_E}{(r_e + R_E) V_{in}}$$

$\Rightarrow A_v = \frac{R_E}{r_e + R_E} \qquad (\because R_E >> r_e)$

$\therefore A_v \simeq 1$.

4) Current gain $(A_I) = \frac{I_o}{I_I} = \frac{I_e}{I_b} = \frac{i_c + i_b}{i_b} = (\beta + 1)$